

Penetration Test Report

AI-Powered Windows Security Assessment

HIGH RISK

Target IP	172.20.10.5
Hostname	N/A
OS Detected	Windows
Scan Date	March 28, 2026 — 16:05
Duration	12.06 seconds
Open Services	3
CVEs Matched	16
Risk Score	7.5 / 10 — HIGH RISK

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01 — Executive Summary

This assessment identified **3 open service(s)** on the target system and matched **16 CVE vulnerabilities** from the local Windows CVE database. Overall risk: **HIGH RISK**. Risk score: **7.5/10**.

Of the 16 matched CVEs, 8 are recent (post-2020) and may be unpatched, while 8 are older CVEs likely patched on modern Windows systems.

Severity	Count	Priority
CRITICAL	3	Immediate action required
HIGH	8	Remediate within 24-48 hours
MEDIUM	4	Remediate within 30 days
LOW	1	Remediate when possible

AI Assessment Overview

Executive Summary The target, a Windows host with open SMB and RPC services (ports 135/tcp and 445/tcp), is vulnerable to several critical and high-severity vulnerabilities listed in the local Windows vulnerability database. The most critical vulnerabilities are CVE-2022-45141 and CVE-2017-0021, bo...

02 — Scan Information

Target IP	172.20.10.5
Hostname	N/A
OS Detected	Windows
Scan Date	2026-03-28 16:05:09
Duration	12.06 seconds
Scanner	Nmap 7.98
Command	/usr/lib/nmap/nmap --privileged -vvv -p 135,139,445 -4 -sV -sC --script smb-vuln...

03 — Open Ports & Services

Port	Protocol	Service	Product / Banner
135	TCP	RPC	Microsoft Windows RPC
139	TCP	SMB	Microsoft Windows netbios-ssn
445	TCP	SMB	microsoft-ds?

04 — CVE Vulnerability Findings

CVE matches are based on service detection from the local NIST NVD dataset. Patch status is estimated by publication date.

Service: 135/rpc

CVE ID	CVSS	Severity	Description	Patch Status
CVE-2017-0100	7.8	HIGH	A DCOM object in Helppane.exe in Microsoft Windows 7 SP1; Windows Server 2008 R2; Windows ...	Likely Patched

CVE-2020-1113	7.5	HIGH	A security feature bypass vulnerability exists in Microsoft Windows when the Task Schedule...	May be Unpatched
CVE-2017-0298	7.3	HIGH	A DCOM object in Helppane.exe in Microsoft Windows Server 2008 SP2 and R2 SP1, Windows 7 S...	Likely Patched
CVE-2018-8514	5.5	MEDIUM	An information disclosure vulnerability exists when Remote Procedure Call runtime improper...	Likely Patched
CVE-2018-8407	5.5	MEDIUM	An information disclosure vulnerability exists when "Kernel Remote Procedure Call Provider...	Likely Patched

Service: 139/smb

CVE ID	CVSS	Severity	Description	Patch Status
CVE-2022-45141	9.8	CRITICAL	Since the Windows Kerberos RC4-HMAC Elevation of Privilege Vulnerability was disclosed by ...	May be Unpatched
CVE-2017-0021	9.0	CRITICAL	Hyper-V in Microsoft Windows 10 1607 and Windows Server 2016 does not properly validate vS...	Likely Patched
CVE-2015-2474	9.0	CRITICAL	Microsoft Windows Vista SP2 and Server 2008 SP2 allow remote authenticated users to execut...	Likely Patched
CVE-2017-0144	8.8	HIGH	The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo...	Likely Patched
CVE-2017-0145	8.8	HIGH	The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo...	Likely Patched
CVE-2025-33073	8.8	HIGH	Improper access control in Windows SMB allows an authorized attacker to elevate privileges...	May be Unpatched
CVE-2023-21549	8.8	HIGH	Windows SMB Witness Service Elevation of Privilege Vulnerability	May be Unpatched
CVE-2022-24500	8.8	HIGH	Windows SMB Remote Code Execution Vulnerability	May be Unpatched
CVE-2025-48802	6.5	MEDIUM	Improper certificate validation in Windows SMB allows an authorized attacker to perform sp...	May be Unpatched
CVE-2021-33783	6.5	MEDIUM	Windows SMB Information Disclosure Vulnerability	May be Unpatched

05 — AI Analysis & Recommendations

Analysis generated by Qwen2.5:3b AI model using RAG from local CVE dataset.

Executive Summary

The target, a Windows host with open SMB and RPC services (ports 135/tcp and 445/tcp), is vulnerable to several critical and high-severity vulnerabilities listed in the local Windows vulnerability database. The most critical vulnerabilities are CVE-2022-45141 and CVE-2017-0021, both of which have a CVSS score of CRITICAL (9.8) and CRITICAL (9.0), respectively.

Vulnerability Analysis

Ranking by exploitability and impact:

1. CVE-2022-45141 (CRITICAL - 9.8):

- Impact: Since the Windows Kerberos RC4-HMAC Elevation of Privilege Vulnerability was disclosed, this vulnerability remains unpatched.
- Description: This vulnerability allows an attacker to elevate privileges over a network.

2. CVE-2017-0021 (CRITICAL - 9.0):

- Impact: This is a Hyper-V system data structure corruption issue that can lead to arbitrary code execution.
- Description: Hyper-V in Windows 1607 and Server 2016 does not properly validate vSMB packet data.

3. CVE-2025-48802 (HIGH - 8.8):

- Impact: An attacker can perform spoofing over a network.
- Description: Improper certificate validation in Windows SMB allows for spoofing.

4. CVE-2021-33783 (HIGH - 8.8):

- Impact: An unauthorized attacker can perform tampering over a network.
- Description: This vulnerability leads to information disclosure in the context of Windows SMB.

5. CVE-2025-59280 (LOW - 3.1):

- Impact: Unauthorized attackers can perform tampering over a network.
- Description: Improper authentication in Windows SMB Client allows for tampering.

Attack Vectors

Most Realistic Attack Paths:

CVE-2022-45141 (CRITICAL - 9.8)

- Exploit Method: Exploit the Kerberos RC4-HMAC Elevation of Privilege Vulnerability.
- Steps:

1. Identify a user with insufficient privileges to access sensitive data or services.

2. Craft a malicious Kerberos ticket that exploits the vulnerability in Windows SMB.

3. Use this ticket to elevate privileges over the network, gaining elevated permissions.

CVE-2017-0021 (CRITICAL - 9.0)

- Exploit Method: Exploit Hyper-V system data structure corruption issue.
- Steps:

1. Identify a user with administrative rights on the target machine.

2. Craft a malicious vSMB packet that exploits the vulnerability in Hyper-V.

3. Execute arbitrary code as an administrator, leading to elevated privileges.

Recommended Tools & Techniques

Metasploit Modules

- CVE-2022-45141: msfvenom for crafting a malicious Kerberos ticket and exploit/windows/smb/ms19_070_sampe for exploiting the vulnerability.
- CVE-2017-0021: Use msfvenom to craft a malicious vSMB packet and exploit/windows/hyper-v/vhdx_exec to exploit Hyper-V.

Other Tools

- Burp Suite: For intercepting and manipulating SMB traffic.
- PowerSploit: For conducting Windows-specific attacks, including elevation of privileges.

Remediation Steps

Specific Patches and Fixes:

1. CVE-2022-45141 (CRITICAL - 9.8):

- Patch Status: Unpatched.
- Remediation: Apply the latest Windows updates, including Microsoft Security Bulletin MS19-070.

2. CVE-2017-0021 (CRITICAL - 9.0):

- Patch Status: Unpatched.
- Remediation: Apply the latest Windows updates, including Microsoft Security Bulletin MS16-053 and MS16-084.

Microsoft KB Numbers:

- CVE-2022-45141: [KB4573691](https://support.microsoft.com/en-us/kb/4573691)
- CVE-2017-0021: [KB4085882](https://support.microsoft.com/en-us/kb/4085882)

Risk Score

Overall Score: 9.5

Justification:

The target is highly vulnerable to critical vulnerabilities, with two of the most severe CVEs (CVE-2022-45141 and CVE-2017-0021) remaining unpatched. The presence of these vulnerabilities coupled with their high CVSS scores indicates a significant risk to the system's security posture. While some other vulnerabilities are rated as HIGH or MEDIUM, they do not significantly alter the overall critica

Conclusion

The Windows host is highly vulnerable to multiple critical and high-severity vulnerabilities that could lead to privilege escalation and data exfiltration. Immediate patching of these vulnerabilities is imperative to mitigate the risk.

06 — Remediation Steps

RPC	Restrict RPC via firewall. Apply all Windows security patches. Disable unused RPC services.
SMB	Disable SMBv1. Apply MS17-010 patch (KB4012212). Block port 445 externally. Enable SMB signing.

General Security Hardening

- Apply all pending Windows security updates via Windows Update
- Enable Windows Defender and keep definitions up to date
- Implement network segmentation and strict firewall rules
- Use principle of least privilege for all accounts
- Enable Windows Event Logging and monitor for suspicious activity
- Implement multi-factor authentication (MFA) for all remote access

07 — Risk Score

Overall Risk Score	7.5 / 10
Risk Level	HIGH RISK
Total CVEs	16
Critical	3
High	8
Medium	4
Low	1
Recent CVEs	8
Likely Patched	8

Risk score is calculated based on recent CVEs (post-2020) only. Older CVEs are weighted lower as they are likely patched on modern Windows systems.

08 — Legal Disclaimer

This report was prepared exclusively for the authorized recipient. All testing was conducted with explicit written authorization from the system owner.

Unauthorized access to computer systems is a criminal offence under the Computer Misuse Act 1990 (UK), Computer Fraud and Abuse Act (USA), and equivalent legislation worldwide.

CVE data sourced from the National Vulnerability Database (NVD) — NIST (<https://nvd.nist.gov/>). AI analysis by Qwen2.5:3b via Ollama (RAG mode — local dataset only).

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